DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Lab Assignment: 22

Aim:
To write a Python program demonstrating string methods such as `upper()`, `lower()`, `split()`, `join()`, `count()`, `replace()`, and `find()`.

Objective:
- To understand various built-in string methods in Python.
- To perform different string manipulations using these methods.
- To analyze how strings can be modified and searched.

Theory:
Strings in Python provide many built-in methods for performing operations such as converting case, splitting text, replacing characters, and searching substrings.
Some commonly used string methods are:
1. **upper()** – Converts all characters of the string into uppercase.
2. **lower()** – Converts all characters of the string into lowercase.
3. **split()** – Splits the string into a list of words based on spaces or a specified delimiter.
4. **join()** – Joins elements of a list into a single string using a specified separator.
5. **count()** – Returns the number of times a specified word or character appears in the string.
6. **replace()** – Replaces occurrences of a word or character with another word or character.
7. **find()** – Returns the index of the first occurrence of a word or character. If not found, it returns **-1**.

These methods help in efficient string manipulation and processing in Python programs.

Algorithm:

1. Start the program.
2. Accept a string from the user.
3. Convert the string to uppercase using `upper()` and display it.
4. Convert the string to lowercase using `lower()` and display it.
5. Split the string into words using `split()` and display the list.
6. Join the list of words using `join()` with – as a separator.
7. Ask the user to enter a word to count and display its occurrence using `count()`.
8. Ask the user to enter a word to replace and the new word.
9. Replace the word using `replace()` and display the result.
10. Ask the user to enter a word or character to find.
11. Use `find()` to display its index.
12. End the program.

Pseudo Code:

```
START

INPUT string

PRINT string in uppercase
PRINT string in lowercase

SPLIT string into words
PRINT list of words

JOIN words using "-"
PRINT joined string

INPUT word to count
PRINT number of occurrences

INPUT word to replace
INPUT new word
PRINT replaced string

INPUT word/character to find
PRINT index using find()

END
```

Code:

```
text = input("Enter a string: ")

# upper()
print("Upper:", text.upper())

# lower()
print("Lower:", text.lower())

# split()
words = text.split()
print("Split:", words)

# join()
joined = "-".join(words)
print("Join:", joined)

# count()
word = input("Enter word to count: ")
print("Count:", text.count(word))

# replace()
old = input("Enter word to replace: ")
new = input("Enter new word: ")
print("Replace:", text.replace(old, new))

# find()
ch = input("Enter word/character to find: ")
print("Find index:", text.find(ch))
```

Output:

```
Enter a string:  hello world
Upper: HELLO WORLD
Lower: hello world
Split: ['hello', 'world']
Join: hello-world
Enter word to count:  l
Count: 3
Enter word to replace:  hello
Enter new word:  hi
Replace: hi world
Enter word/character to find:  e
Find index: 1
```

Conclusion:

Thus, the Python program was successfully executed to demonstrate various **string methods** such as `upper()`, `lower()`, `split()`, `join()`, `count()`, `replace()`, and `find()` for performing different string manipulation operations.